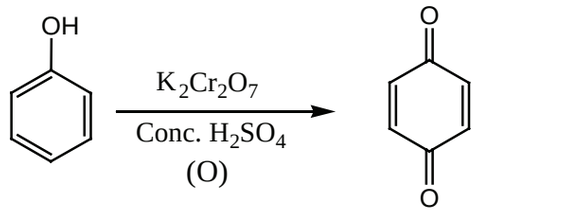



Phenol Sodium Phenoxide Anisole (methoxybenzene)

### d) Oxidation:

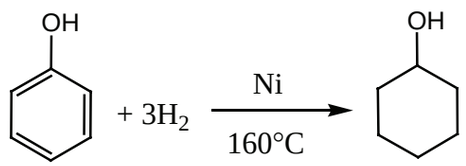
Phenol undergoes oxidation with air or acidified $K_2Cr_2O_7$ with conc. $H_2SO_4$ to form 1,4-benzoquinone.




Phenol 1,4-benzoquinone

### e) Reduction:

Phenol on catalytic hydrogenation gives cyclohexanol.




Phenol cyclohexanol

## Reactions of benzene ring:

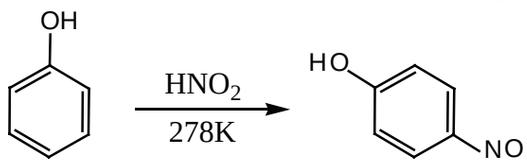
### Electrophilic aromatic substitution:

We have already learnt in XI standard that the groups like -$\ddot{O}H$, -$\ddot{N}H_2$, etc., which when directly attached to the benzene ring, activate the ring towards electrophilic substitution reaction and direct the incoming electrophile to occupy either the ortho or para position.

**Common electrophilic aromatic substitutions are as follows:**

### i) Nitrosation:

Phenol can be readily nitrosoated at low temperature with $HNO_2$.




phenol p-nitroso phenol



XII U11-Hydroxy compounds.indd 128 11-01-2021 15:44:57